vl4dio4n at #SMM4H-HeaRD 2026 Task 7: Comparing Encoder-, LLM-, and SVM-Based NER Pipelines for Substance-Use Impact Detection

Anonymous ACL submission

Abstract

We describe our submission to SMM4H-HeaRD 2026 Task 7, which asks systems to label `ClinicalImpacts` and `SocialImpacts` spans in Reddit posts about non-medical substance use. We compare four pipeline shapes built on the same DeBERTa-v3-base backbone: (i) a direct 5-class encoder with a linear-chain CRF head, (ii) a two-stage detect-then-classify pipeline that delegates span typing to an instruction-tuned LLM (Qwen2.5-7B or Gemma-3-12B, 4-bit NF4), (iii) an audit pipeline in which the same LLM verifies the encoder's predictions, and (iv) a classical-ML variant that replaces the LLM with an SVM trained on encoder span embeddings. Across 16 configurations, the encoder-only *DeBERTa-v3 + CRF* configuration is the strongest single system on the official test split, reaching 45.4% strict and 54.2% relaxed F1 — +8.6 / +5.3 points above a `mental-roberta-base` baseline. LLM audits give a small dev gain that does not transfer to test.

1 Introduction

SMM4H-HeaRD 2026 Task 7 (Dey et al., 2025) releases a corpus of Reddit posts about non-medical opioid use, annotated under a 5-label BIO scheme that distinguishes `ClinicalImpacts` (e.g. *withdrawal*, *rehab*) from `SocialImpacts` (e.g. *lost my job*, *homeless*). Systems are scored with strict micro-F1 and a relaxed overlap F1 in the CoNLL-2003 style (Tjong Kim Sang and De Meulder, 2003), computed by `seqeval`.

Our submission is a comparative study, not a single architectural proposal. We fix the encoder backbone to `DeBERTa-v3-base` (He et al., 2023) and vary what sits on top of it, asking two questions. **RQ1.** Does decoupling span detection from span typing improve over a well-decoded 5-class CRF? **RQ2.** Does adding an LLM or SVM auditor on top of the CRF recover any appreciable F1?

We answer both questions with a 16-row ablation (Table 1). The encoder-only DeBERTa-v3 + CRF configuration is the strongest *single* system on the test split, and none of the LLM- or SVM-augmented variants improves on it. A 4-shot Gemma-3-12B audit obtains the best dev strict F1 but loses to the encoder on test, which we attribute to the distributional shift between dev and test on this corpus.

2 Data

The released corpus consists of 842 training posts (∼17.2K tokens), 258 development posts (∼5.2K tokens), and 278 test posts (∼6.2K tokens). The training and development CSVs contain (mostly) one row per post; the test CSV is aggressively segmented into 1.7K sentence-level rows, which forces inference to run on shorter contexts than training. The 5 BIO labels are heavily imbalanced: the `O` tag accounts for ∼94% of all training tokens (a ∼16:1 ratio against entity tokens). On train and dev, `ClinicalImpacts` spans outnumber `SocialImpacts` spans by ∼3:1, but on the test set the ratio inverts and `SocialImpacts` becomes the majority class — a shift we revisit in Sec. 5. The test set is also noticeably more first-person (∼71% of segments) than train or dev (∼45%).

3 System Description

We compare four pipelines, all built around the same DeBERTa-v3 backbone fine-tuned with a maximum sequence length of 256, batch size 16, 20 epochs, 10% linear warm-up and a base learning rate of 2×10^{-5}.

Direct + CRF (Exp 04, our main system). The encoder predicts 5-class BIO tags directly. A linear-chain CRF (Lafferty et al., 2001; Lample et al., 2016) on top of the encoder replaces token-level cross-entropy with the CRF negative log-likelihood and decodes via Viterbi at inference. The CRF rules out illegal BIO transitions, which mainly cleans up boundaries (and hence strict F1) without changing the set of detected entity regions. We add Layer-wise Learning-Rate Decay (Howard and Ruder, 2018) with multiplier 0.9 to stabilise DeBERTa fine-tuning. This configuration is our final Codabench submission.

Two-stage with LLM span typer (Exp 05a, 07a, 08a, 09a). The encoder is re-trained on a collapsed 3-class label set ({`O`, `B-Imp`, `I-Imp`}) and only detects span boundaries; an instruction-tuned LLM types each predicted span as `ClinicalImpacts`, `SocialImpacts`, or `Neither`. We use focal loss (Lin et al., 2017) (γ=2) to handle class imbalance in the detector, since the CRF is no longer in charge of the 5-class structure.

Audit with LLM verifier (Exp 05b, 07b, 08b, 09b). The 5-class DeBERTa-v3 + CRF detector of Exp 04 produces a full BIO tagging; the LLM is shown the encoder's predicted span and label as a hint and may verify, correct (Clinical $\leftrightarrow$ Social), or reject it (mark as Neither, which removes the span). Audit and two-stage share the same prompt template and class descriptions; only the candidate construction differs.

LLM details. We use `Qwen/Qwen2.5-7B-Instruct` and `google/gemma-3-12b-it`, loaded in 4-bit NF4 precision via `bitsandbytes` (Dettmers et al., 2023) so that both fit on a single consumer GPU. Decoding is greedy and capped at 6 new tokens; the post is truncated to 2,000 characters and the candidate span to 200. We compare zero-shot prompting with 4-shot in-context learning (Brown et al., 2020); few-shot demonstrations are mined from gold training spans and balanced across the three classes (one per class plus one randomly drawn `Neither`).

SVM-augmented variant (Exp 10a/10b). To isolate how much of the LLM's contribution comes from world knowledge that is not already linearly accessible from the encoder, we replace the LLM with an SVM (Cortes and Vapnik, 1995). Each candidate span is represented by the mean-pooled hidden state of its tokens in the (frozen) fine-tuned encoder; a `StandardScaler` $\rightarrow$ `PCA(128)` $\rightarrow$ `SVC(RBF, class_weight=balanced)` pipeline is fitted with 5-fold grid-searched macro-F1 over C and γ.

Baseline. For comparability we keep the `mental-roberta-base` (Ji et al., 2022) + cross-entropy, 5-class direct configuration as our internal baseline (Exp 01); it is a RoBERTa (Liu et al., 2019) variant adapted to mental-health Reddit text and is the natural starting point given the domain.

4 Experiments

We report the full 16-row ablation in Table 1. Each row pairs a pipeline shape (Pipe: `D` direct, `2S` two-stage, `Au` audit, `Ens` ensemble, `2S/Au+SVM` SVM-augmented two-stage/audit), a backbone, a loss function, and an external span classifier when applicable. "Dev" columns are computed on the official 258-row development split with `seqeval`; "Test" columns are the official Codabench leaderboard scores returned for the corresponding submission.

5 Results and Discussion

Encoder choices and CRF dominate. Switching the backbone from `mental-roberta-base` (Exp 01) to `deberta-v3-base` with focal loss and LLRD (Exp 03) is the single largest step on test relaxed F1 (+6.9 over Exp 01). Adding the CRF head (Exp 04) gives a further +3.0 test strict points and reduces the strict-relaxed gap from 19.4 pts (Exp 03, dev) to 8.8 pts on test, confirming that the CRF mainly cleans up span boundaries rather than discovering new entities. Exp 04 is the only configuration that exceeds 0.45 test strict F1 across the entire ablation.

LLM auditing helps on dev, not on test. The strongest LLM configuration is *audit + Gemma-3-12B + 4-shot* (Exp 09b), which reaches 0.426 dev strict F1 — the highest in the table — but only 0.432 test strict, $\sim$2 points below Exp 04. The two-stage shapes consistently lose to the audit shapes at relaxed F1 because the 3-class detector finds slightly worse spans than the 5-class CRF and the LLM cannot fix upstream boundary errors. Within prompting regimes, 4-shot in-context demonstrations help Gemma (e.g. +3.8 dev strict on the audit pipeline) but slightly hurt Qwen ($-$0.6 dev strict), suggesting that the larger model extracts more signal from demonstrations.

SVM auditing is essentially neutral. The audit SVM (Exp 10b) preserves the Exp 04 dev numbers (0.407 strict / 0.559 relaxed) and reaches a 5-fold macro-F1 of 1.0 on gold training embeddings: the two impact classes are linearly separable in the

Exp	Pipe	Backbone	Loss	CRF	External classifier	Dev S	Dev R	Test S	Test R
01	D	mental-roberta-base	CE	N	—	0.356	0.489	0.368	0.489
02a	D	mental-roberta-base	W-CE	N	—	0.360	0.493	0.410	0.549
02b	D	mental-roberta-base	Focal	N	—	0.339	0.492	0.336	0.507
03	D	deberta-v3-base	Focal	N	—	0.345	0.539	0.424	**0.558**
04	D	deberta-v3-base	CRF NLL	Y	—	0.413	**0.571**	**0.454**	0.542
06	Ens	deberta-v3-base ×3	CRF NLL	Y	—	0.410	0.565	0.441	0.551
05a	2S	deberta-v3-base (3-cls)	Focal	N	Qwen2.5-7B, zero-shot	0.394	0.423	0.421	0.449
05b	Au	deberta-v3-base	CRF NLL	Y	Qwen2.5-7B, zero-shot	0.386	0.541	0.443	0.549
07a	2S	deberta-v3-base (3-cls)	Focal	N	Qwen2.5-7B, 4-shot	0.381	0.431	0.434	0.483
07b	Au	deberta-v3-base	CRF NLL	Y	Qwen2.5-7B, 4-shot	0.380	0.537	0.440	0.552
08a	2S	deberta-v3-base (3-cls)	Focal	N	Gemma-3-12B, zero-shot	0.382	0.421	0.395	0.418
08b	Au	deberta-v3-base	CRF NLL	Y	Gemma-3-12B, zero-shot	0.388	0.550	0.400	0.496
09a	2S	deberta-v3-base (3-cls)	Focal	N	Gemma-3-12B, 4-shot	0.408	0.438	0.428	0.444
09b	Au	deberta-v3-base	CRF NLL	Y	Gemma-3-12B, 4-shot	**0.426**	0.567	0.432	0.523
10a	2S+SVM	deberta-v3-base (3-cls)	Focal	N	SVM (RBF, PCA-128)	0.357	0.367	0.433	0.435
10b	Au+SVM	deberta-v3-base	CRF NLL	Y	SVM (RBF, PCA-128)	0.407	0.559	—	—

Table 1: Ablation across all 16 configurations. Pipelines: `D` = direct, `2S` = two-stage, `Au` = audit, `Ens` = ensemble, `2S/Au+SVM` = SVM-augmented. "Dev" and "Test" columns give strict (S) and relaxed (R) F1 on the development split and the Codabench test split, respectively. Layer-wise LR Decay is enabled for all DeBERTa-v3 runs (Exp 03 and above). Best per metric in **bold**.

CRF encoder's feature space, so the SVM rarely overrides the encoder. This is the cleanest evidence in our ablation that the LLM's added value over the CRF is bounded by the linear information already present in the encoder representation.

Recall is the bottleneck. Across the entire ablation, relaxed precision climbs from 0.58 (Exp 01) to 0.73 (Exp 09b) on dev, while relaxed recall stays in the 0.42–0.48 band. Our systems become progressively more conservative but do not discover materially more gold spans. Combined with the train/dev vs. test class flip (`ClinicalImpacts` majority on train/dev, `SocialImpacts` majority on test), this explains why the dev-best configuration (Exp 09b) is not the test-best configuration (Exp 04).

Comparison to the organisers' baseline. Our final submission (Exp 04) is ∼7 relaxed-F1 points behind the DeBERTa-large baseline of ∼0.61 relaxed F1 reported by Dey et al. (2025). The most plausible drivers of the gap are: (i) we use the *base* variant of every backbone we tried, (ii) our pipeline processes each released CSV row independently and therefore cannot recover entities that span multiple sentence-level test segments, and (iii) our 4-shot demonstrations are sampled by class only, with no balancing for narrative perspective or span length.

6 Conclusion

We presented a controlled comparison of four NER pipeline shapes on SMM4H-HeaRD 2026 Task 7. The strongest single system on the official test split is a 5-class DeBERTa-v3 + CRF encoder (Exp 04, 0.454 strict / 0.542 relaxed F1). LLM and SVM auditing on top of the same encoder bring at most a small dev gain that does not transfer to test, mostly because the test split shifts to a more first-person, `Social`-heavy distribution that the class-only 4-shot sampler cannot anticipate. Recall on rare `SocialImpacts` expressions remains the dominant failure mode, suggesting that a larger backbone, span-merging across sentence-level test segments, and perspective-aware in-context retrieval are the most promising directions for future work.

Limitations

We use only 4-shot in-context demonstrations and do not fine-tune the LLMs on the auditing task; parameter-efficient fine-tuning on the training spans may close the gap to the organisers' baseline. Our SVM head is restricted to linear-RBF kernels with 128 PCA components; richer span representations (e.g. concatenation with first/last token embeddings) were not explored. Each released CSV row is processed independently, so our pipeline cannot recover impact spans that cross sentence-level test segments.

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